

CADENCE: A prospective, leakage-sealed test of cross-animal causal dynamical transfer

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Pre-outcome manuscript framework — 25 July 2026

Pre-outcome status. This manuscript specifies a prospective analysis whose primary claim may fail. No public-seed teacher evaluation-world result, Allen evaluation-mouse response, or ICMS target response is reported here. Every empirical result slot is marked NOT EVALUATED and must remain so until its predeclared post-freeze aggregation; outcome-sequestered biological cohorts additionally require authorized unsealing.

Abstract

Cross-animal alignment can reveal shared observational dynamics, but it does not by itself show that a causal response operator transports to a new animal. We introduce CADENCE, a hierarchical controlled state-space model and a prospective test of a deliberately stronger claim: after an intervention operator is learned from donor animals, can the complete neural and behavioral response of a held-out animal be predicted when that animal contributes only unperturbed activity? The model separates shared nonlinear normal flow, a state-dependent low-rank intervention field, capacity-limited recording-adaptor residuals, adaptor-specific observation maps, and a shared behavior readout. Its stage-locked biological protocol physically separates target inputs from target post-onset outcomes and hashes open-loop prediction bundles before a single unsealing step. The primary biological test uses 28 sequestered mice from the Allen Visual Behavior Ophys release; a six-mouse intracortical microstimulation analysis is explicitly exploratory, with a randomized causal contrast available in five mice. A public-seed procedural teacher benchmark tests recoverability under known shared operators and supports a separate impossibility construction, but is ineligible for a headline conjunction. We state the latent gauge invariance and the stronger observation-map non-identifiability that arises when perturbations leave target normal support. Results will be interpreted through a prespecified eight-part biological conjunction; failure or missing evidence on any required component precludes the headline. This version is a methods and analysis framework, not a report of evaluation performance.

1 Introduction

Neural population recordings differ across animals in cell identity, measurement geometry, and sampling, even when behavior and task structure are shared. A substantial literature therefore aligns latent representations across sessions and subjects, transfers decoders, or identifies conserved population dynamics (Chen et al., 2015; Dyer et al., 2017; Gallego et al., 2020; Safaie et al., 2023; Schneider et al., 2023; Gosztolai et al., 2025; Jiang et al., 2025). Latent dynamical systems provide complementary machinery for smooth single-trial inference, nonlinear generation, switching flow, and behaviorally prioritized prediction (Yu et al., 2009; Pandarinath et al., 2018; Linderman et al., 2017; Glaser et al., 2020; Sani et al., 2024). These achievements make cross-animal causal prediction plausible; they do not establish it.

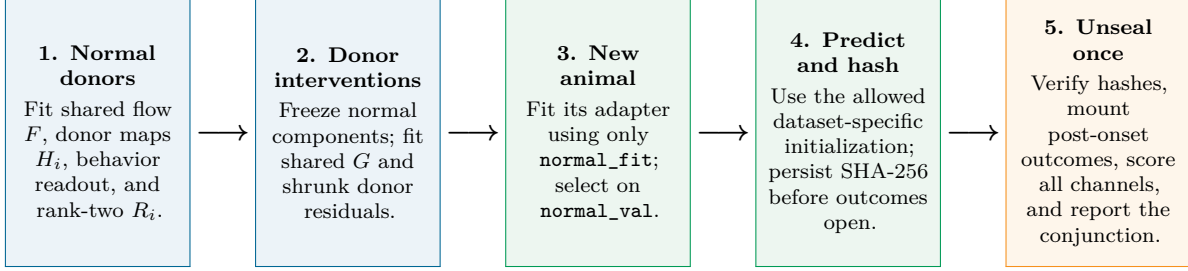


Figure 1: **The information boundary is part of the estimand.** Target-animal intervention outcomes cannot affect preprocessing, adaptation, early stopping, uncertainty calibration, or method selection.

The distinction is consequential. An observational alignment can predict where ordinary trajectories lie while being wrong about the vector field that moves a trajectory after intervention. Conversely, an intervention operator may be shared in a latent coordinate system while the target animal’s neural readout is underdetermined precisely where the intervention sends the state. Perturbation is therefore not merely another test condition. It probes model structure outside the information used to register a new animal (Shenoy and Kao, 2021; Galgali et al., 2023).

The closest adjacent biological result fits constrained cortical networks to unperturbed activity and evaluates out-of-distribution optogenetic inactivation (Sourmpis et al., 2026). That work demonstrates why normal fit is insufficient and why perturbation tests matter. Its in-vivo perturbation endpoint is behavior, however, rather than a simultaneously measured full neural response in a named held-out animal registered from normal activity alone. CADENCE targets this narrower unresolved experiment. It does not claim that intervention prediction, cross-animal alignment, or neural dynamical modeling is itself new.

Our proposed contribution is conditional:

If every frozen gate passes, CADENCE will provide a leakage-free demonstration that a donor-trained intervention operator can predict full neural and primary-behavior trajectories in a held-out animal whose mapping was learned only from normal activity.

“Unseen” means unseen in the target animal. The intervention family and its physical descriptor are observed in donors; globally novel intervention classes are outside the claim. A positive sensory-omission result would not establish transfer of direct neural control, and the small ICMS cohort remains exploratory regardless of how favorable its evaluable predictive metrics are.

2 Formal target of inference

2.1 Units, information sets, and potential trajectories

Let i index animals, q index eligible intervention queries, t index post-onset time, and a denote an intervention descriptor. The multivariate potential trajectory

$$Y_{iq}(t; a) = (Y_{iq1}^{\text{neural}}(t; a), \dots, Y_{iqp_i}^{\text{neural}}(t; a), Y_{iq}^{\text{behavior}}(t; a))$$

contains every prespecified neural channel and primary behavioral variable. The target-animal effect is

$$\tau_i(t, a) = \mathbb{E}_q[Y_{iq}(t; a) - Y_{iq}(t; 0) \mid i, q \text{ eligible}]. \quad (1)$$

The expectation is over the dataset-specific eligible risk set, not over a hypothetical unrestricted intervention population.

For target animal i^* , fitting has access to donor normal and intervention data and target normal support $\mathcal{D}_{i^*}^{\text{normal}}$. Prediction uses scheduled inputs, intervention descriptors, and the dataset-specific initialization described below. It has no access to target post-onset neural activity, animal-generated action/choice, reward, success label, or response time. Deterministic query construction may use only the predeclared, allowed schedule, descriptor, and initialization; target intervention outcomes may not enter fitted or response-dependent preprocessing, normalization, channel selection, early stopping, hyperparameter selection, or uncertainty calibration. The predicted causal trajectory is the difference between treated and control open-loop rollouts from the same encoded initialization:

$$\hat{\tau}_{i^*q}(t, a) = \hat{Y}_{i^*q}(t; a) - \hat{Y}_{i^*q}(t; 0).$$

The animal is the biological replication unit. Trials, cells, conditions, sessions, time bins, and random optimization seeds are repeated measurements within animal and cannot increase the inferential sample size. In the procedural benchmark, the independently generated world is the unit: its four targets are averaged equally before any bootstrap or sign-flip test. These world-level intervals remain procedural and non-headline.

2.2 Dataset-specific causal contrasts

Allen omissions occur in an eligible expected-flash risk set. Within each mouse, the omitted-flash trajectory is contrasted with a separate normal effect-control pool. Controls are chosen without reading outcomes by the frozen hierarchy: exact preceding image and flashes-since-change; image plus a prespecified risk bin; image only; risk bin only; then the complete eligible pool. Fallback and effective control counts are reported. The corresponding estimand is therefore an eligible-risk-set omission effect, not an unqualified effect of arbitrary visual removal.

For ICMS, each positive-current condition is contrasted within session and trial block against randomized zero-current catches. Conditions are first aggregated within session and sessions are then weighted equally within animal. Five mice are design-eligible; the primary aggregate additionally requires every prespecified session/condition to have validated same-block catch support, or the five-mouse cohort is incomplete and NOT EVALUATED. ICMS83 has no released catch trials; its guarded-ITI reference supports normal calibration and a nonrandomized sensitivity, but it cannot enter the causal-skill conjunction.

Allen and teacher rollouts start from each query’s final pre-onset sample. ICMS is a prespecified marginal condition–response exception: before target stimulation metadata are opened, each session’s predictions are averaged over 16 signal-blind normal-audit contexts on the full physical action lattice. Scoring later interpolates those curves to realized descriptors and compares condition means. ICMS therefore does not test trial-specific state initialization.

3 Model

3.1 Hierarchical controlled dynamics

CADENCE posits a latent state $z_{it} \in \mathbb{R}^d$, scheduled task input u_{it} , and action vector a_{it} . Here i indexes a recording adapter in the model equations: an animal for teacher/Allen and an animal–

session pair for ICMS; inferential replication remains at the animal level.

$$z_{i,t+1} = F_\theta(z_{it}, u_{it}) + R_i(z_{it}) + G_\phi(z_{it})a_{it} + \delta_i(a_{it}), \quad (2)$$

$$y_{it}^{\text{neural}} = H_i^{\text{neural}}(z_{it}) + \epsilon_{it}^y, \quad (3)$$

$$y_{it}^{\text{behavior}} = A_i H^{\text{behavior}}(z_{it}) + b_i + \epsilon_{it}^b. \quad (4)$$

The shared normal transition is a residual discretization

$$F_\theta(z, u) = z + \Delta\{-z + f_\theta([z, u])\}.$$

The recording-adaptor normal residual is capacity limited:

$$R_i(z) = \Delta U_i V_i \tanh(z), \quad \text{rank}(U_i V_i) \leq 2.$$

For intervention channel c , the implemented state-conditioned field is

$$g_c(z) = \Delta \left[b_c + L_c M_c \tanh(z) \right], \quad \text{rank}(L_c M_c) \leq r_G, \quad (5)$$

and $G_\phi(z)a = \sum_c a_c g_c(z)$. The full teacher configuration uses $d = 8$, $r_G = 2$, and $\Delta = 0.05$; full Allen uses $d = 12$, $r_G = 2$, and $\Delta = 0.1$; and full ICMS uses $d = 12$, $r_G = 4$, and $\Delta = 1/30$. All three use rank-two adapter residuals. A shrunk additive donor term δ_i , grouped by animal, captures residual intervention heterogeneity. It is never estimated for a target: the point prediction uses its donor-distribution mean of zero, while uncertainty draws integrate over the donor distribution. Training-donor deltas are projected to exact zero mean after every optimizer step, including after the final update.

Each recording adapter has an encoder, neural decoder, and affine behavior calibration because recorded cells are not matched. The adapter is animal-specific for teacher and Allen and session-specific (but nested within animal) for ICMS. Donor intervention residuals remain grouped by animal. Behavior otherwise uses a shared decoder. Neural and behavioral observations are jointly encoded during normal-only adaptation; future target samples are never teacher-forced into an intervention rollout. For the teacher benchmark’s primary neural endpoint, every learned method uses a separate softplus-Poisson linear quasi-likelihood readout fit on frozen open-loop target normal-fit rollouts. Adam uses the frozen readout weight decay of 0.2, and the additional explicit ridge coefficient is selected only on normal-validation rollouts. The model’s native neural decoder is retained as a separately labeled diagnostic, not silently substituted into the primary teacher score.

3.2 Stage-locked fitting

The optimization stages instantiate the information boundary in Figure 1:

- Stage 1.** Fit F_θ , the shared behavior decoder, donor observation maps, and donor R_i using normal donor sequences.
- Stage 2.** Freeze normal components. Fit G_ϕ and shrunk donor intervention residuals on training-donor perturbations, with whole-animal validation.
- Stage 3.** Register target i^* . Fit its encoder, observation map, behavior calibration, and capacity-limited residual only on $\mathcal{D}_{i^*}^{\text{normal}}$; select checkpoints using target normal validation.
- Stage 4.** Freeze every parameter. For trial-conditional teacher and Allen queries, encode the final pre-onset sample; for ICMS, use the frozen normal-audit context ensemble. Roll out treated and control schedules open loop, then serialize and hash every prediction.

Stage 5. Verify the hash and freeze attestation, mount the physically separate target outcomes once, and score.

Locked preparation fsyncs an active-seal journal before changing target permissions. Interrupted uncommitted artifacts are quarantined rather than overwritten or deleted, and re-entry idempotently re-seals or restores the target. The journal/registry is removed only after the score completion and a separately hashed post-score restoration completion are durable.

Within each outer fold, candidate F is fit only on intervention-training donors. The validation animal receives normal-only adapters with F frozen and selects G epochs. A fresh model then refits F and G on all outer donors for those fixed epoch counts before target adaptation.

Dataset-specific normal objectives use masked reconstruction, dynamics consistency, variance-collapse control, and residual shrinkage. ICMS additionally uses a variational KL term; teacher adds a short open-loop loss. The intervention objective evaluates complete donor post-onset rollouts after encoding only the last pre-onset sample. Architecture, optimization budget, and all early-stopping data are fixed before sequestered biological outcomes open.

4 What is and is not identifiable

The observed-space causal trajectory in Equation 1 is the scientific target. Absolute latent coordinates are not.

Proposition 1 (Functional latent gauge invariance). *Viewed as unrestricted functions, not necessarily within the implemented factorized-tanh parameter class, let $Q : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be invertible and define $\tilde{z} = Qz$. For any model $(F, G, R_i, \delta_i, H_i)$, define*

$$\begin{aligned}\tilde{F}(\tilde{z}, u) &= QF(Q^{-1}\tilde{z}, u), & \tilde{G}(\tilde{z}) &= QG(Q^{-1}\tilde{z}), \\ \tilde{R}_i(\tilde{z}) &= QR_i(Q^{-1}\tilde{z}), & \tilde{\delta}_i(a) &= Q\delta_i(a), \\ \tilde{H}_i(\tilde{z}) &= H_i(Q^{-1}\tilde{z}).\end{aligned}$$

Then the transformed and original models induce identical observed normal and intervention trajectories.

Proof. If $\tilde{z}_t = Qz_t$, substitution into the transformed transition gives $\tilde{z}_{t+1} = Qz_{t+1}$ by induction. The transformed observation is $\tilde{H}_i(\tilde{z}_t) = H_i(z_t)$. Thus every likelihood and observed-space causal effect is unchanged. $\square$

Consequently, recovery of F or G in a particular coordinate system is not meaningful without gauge alignment. Teacher diagnostics fit an affine gauge using target *normal-fit* states only; they remain diagnostics and cannot substitute for observed neural and behavioral prediction.

Gauge freedom is benign because it leaves observations invariant. A more serious ambiguity appears when the target observation map is learned only on normal support.

Proposition 2 (Normal-support non-identifiability). *Let $\mathcal{S}_i \subset \mathbb{R}^d$ be the support of target normal latent states and let $\mathcal{T}_i$ contain states visited after intervention. Suppose the observation class contains two maps H_i and H'_i that agree on $\mathcal{S}_i$ but differ at some $z^* \in \mathcal{T}_i$. Then no procedure using donor data and target normal data alone can identify the target observation at z^* .*

Proof. The two maps induce exactly the same distribution for every target datum available before unsealing, so any normal-only estimator has the same distribution under both models. Yet their target intervention observations at z^* differ. Therefore the available information cannot distinguish the two predictions. $\square$

Table 1: **Frozen evaluation resources.** Counts describe the prespecified inputs, not model performance. Sessions, trials, and cells are repeated measurements. Animals are the biological replication units; in the teacher benchmark, four targets are averaged within each independently generated world and the world is the inferential unit.

Resource	Independent units	Frozen scope
Teacher RNN	10 development and 20 public-seed procedural evaluation worlds; 4 nested targets per world	Known nonlinear dynamics, six intervention channels, variable 64–128-neuron maps, paired counterfactual twins, and an explicit non-identifiability pair. Internal locked label is retained only for schema compatibility.
Allen Visual Behavior Ophys 1.1.0	32 mice: 4 development and 28 locked evaluation mice	VISp, Slc17a7-IRES2-Cre, 175 μm , familiar active sessions; 6,768 cells, 65,522 clean normal windows, and 4,707 eligible omission windows.
DANDI:001868 v0.260715.2016	6 task mice; randomized causal contrast in 5	45 trimodal source sessions (v1 scores spikes and wheel); 14,734 canonical 100-Hz, 70-pulse trials; 1,400 randomized catches and 932 signal-blind guarded-ITI calibration windows. ICMS83 is absolute-trajectory-only.

The ambiguity exists even for linear observation maps. If normal states span a proper subspace $S \subsetneq \mathbb{R}^d$, choose $v \in S^\perp$ and any nonzero output vector c . Then

$$H'_i(z) = H_i(z) + cv^\top z$$

agrees with H_i on all normal states but differs whenever an intervention produces $v^\top z \neq 0$. Low rank in Equation 5 regularizes the intervention field but does not resolve this ambiguity: a rank-one perturbation can still leave S . Linear H_i becomes estimable—within the shared gauge—only under adequate target normal excitation; a nonlinear H_i additionally requires coverage or structural regularity beyond the observed support.

Corollary 1 (Impossibility pair). *Consider two worlds that have identical donor and target normal distributions but opposite private target intervention effects, τ and $-\tau$. Any normal-only target predictor p is identical in the two worlds, and its average squared error across them is at least that of the zero-effect predictor.*

Proof.

$$\frac{1}{2}\{\|p - \tau\|^2 + \|p + \tau\|^2\} = \|p\|^2 + \|\tau\|^2 \geq \|\tau\|^2.$$

The final term is the zero-effect error in either world. $\square$

The teacher benchmark implements this pair by making the target private intervention direction independent of all normal data and flipping its sign. A model should fail there. More generally, a negative neural result alongside positive latent or behavioral diagnostics is compatible with Proposition 2; it is not evidence that an arbitrary observation map was actually recoverable.

5 Evaluation resources

5.1 Procedural teacher RNN

The teacher generator fixes a nonlinear contractive normal operator, state-dependent rank-two intervention fields, exact rank-two animal residuals, animal-specific 64–128-neuron observation maps, three behaviors, and negative-binomial neural noise. Each intervention trial has a counterfactual

twin with the same initial state, task drive, process innovations, and observation-noise variables. Ten worlds are reserved for development and 20 for a predeclared deterministic post-freeze procedural evaluation, with four nested targets per world. All 20 numeric evaluation seeds are public plaintext in `configs/teacher.yaml`; no seed secrecy, outcome blinding, or sequestration is claimed. The internal partition label `locked` is retained only for schema compatibility. These 30 default seed worlds instantiate the frozen main regime; they do not silently vary stress axes. Target metrics are averaged equally within world, leaving $n = 20$ independent units for procedural intervals. Those intervals are descriptive and can never enter a biological or headline conjunction. Separately, the generator supports explicit diagnostic stress settings for conserved intervention strength, animal residual strength, donor support, target population size, normal-state coverage, and the impossibility construction in Corollary 1. Semantically addressed random streams ensure that explicitly adding such a setting does not alter unrelated examples.

5.2 Allen Visual Behavior Ophys

The primary biological cohort is frozen from the Allen Visual Behavior Ophys 1.1.0 release (Allen Institute for Brain Science, 2021): one familiar active VISp experiment per `Slc17a7-IRES2-Cre` mouse at 175 μm , with at least 40 cells. Four mice selected by a deterministic hash rule are development-only; 28 mice remain in five locked outer folds. The full $[-1, 2]$ -s window must have common neural, running, pupil, and lick support. Normal anchors are active ordinary presentations, at least 3 s from an omission, change, or sham. A deterministic 160-window adapter pool is split 70/15/15% into normal fit, validation, and audit sets; every remaining clean normal forms a separate effect-control pool. The primary outcomes over 0–2 s are every prespecified cell’s event-rate trajectory and running speed. Pupil area and lick rate are secondary and cannot rescue a failed running gate.

5.3 DANDI:001868 ICMS

The exploratory direct-intervention analysis uses the immutable published DANDI:001868 version 0.260715.2016 (Kim et al., 2026b,c,a). Six task mice contribute 45 sessions containing behavior, ephys, and ophys. Each leave-one-animal-out fold uses one rotating whole-animal validation donor. Target normal calibration combines zero-current catches with signal-blind ITI windows selected from the complement of all trial and stimulation intervals after a 2-s guard on both sides. No pre-stimulation fragment of a target intervention trial is relabelled as normal. The source-authored `is_good_trial` flag is an additional prespecified signal-support restriction applied before neural or wheel signals are read. Its released NWB description defines `true` as matching two-photon coverage of the stimulation window, `false` as no overlapping imaging frames, and defaults to `true` when no source MAT key exists. It is not a hit/response filter and is not used for tuning; the ICMS estimand is explicitly conditional on this released cross-modal-coverage flag.

The primary intervention family is the canonical 100-Hz, 70-pulse, 167- μs train (approximately 0.7 s). Longer or otherwise noncanonical recorded trains are outside the frozen query lattice and labeled `OUT_OF_SCOPE_NOT_SCORED`; their analysis is NOT EVALUATED, not a duration-generalization sensitivity. Actions use physical current, pulse physics, and electrode coordinates rather than implant-specific raw channel identifiers. The primary neural endpoint is curated sorted-unit activity with a conservative 2-ms pre-train through 5-ms post-train exclusion. Frozen v1 processed files do not retain a distinct unmasked full-train spike tensor, so the full-train spike analysis is NOT EVALUATED. All source sessions include ephys, ophys, and wheel data, but frozen v1 models and scores use sorted spikes and wheel only. Sparse volumetric z-stack observations remain in processed

files for a future native-mask analysis and are NOT EVALUATED here; they are not described as a simultaneous 30-Hz population.

ICMS responses can depend on current, distance, direct axonal recruitment, and transsynaptic dynamics (Stoney et al., 1968; Tehovnik et al., 2006; Butovas and Schwarz, 2003; Histed et al., 2009; Lycke et al., 2023). This biology motivates a physical action descriptor and cautions against interpreting stimulation as one generic binary input.

6 Scores, uncertainty, and decision rule

6.1 Full-trajectory scores

For target i , let s_{ik} be a channel scale estimated exclusively from that animal’s normal-fit support, and let $m_{iqt k}$ be the fixed observation mask. Causal skill is

$$\text{CS}_i = 1 - \frac{\sum_{q,t,k} m_{iqt k} [\{\hat{\tau}_{iqt k} - \tau_{iqt k}\}/s_{ik}]^2}{\sum_{q,t,k} m_{iqt k} (\tau_{iqt k}/s_{ik})^2}. \quad (6)$$

A perfect prediction scores 1, prediction of no causal effect scores 0, and a negative value is worse than no effect. Every retained time–channel element is scored; responsive-cell selection is forbidden. Scores are first computed within animal and then averaged with equal animal weight. The scale s_{ik} is the sample standard deviation on target normal-fit support, clipped to the within-animal 10th–90th percentiles of positive, finite channel standard deviations. A zero-effect-energy stratum has undefined skill; it is retained in the stratum ledger but excluded from the finite-stratum mean.

Allen’s frozen primary is a dataset-specific exception to the pooled ratio in Equation 6. Within each mouse, predicted and observed effects are first averaged within every eligible image identity $\times$ risk bin $\times$ matching-fallback stratum. Causal skill is computed separately in each stratum and then averaged across strata. The ratio pooled over all query–time–channel elements is retained only as a secondary Allen summary.

Frozen v1 emits normalized trajectory RMSE and selected time-resolved observed-space summaries. It does not emit response-summary or split-half reliability-ceiling artifacts; those fields remain NOT EVALUATED. Absolute-trajectory error is secondary because shared baseline activity can make a failed effect prediction look accurate. A positive claim requires posterior trajectory draws evaluated by the multivariate energy score, which is proper and lower-is-better. If a runner does not emit valid draws and this score, the proper-score gate remains NOT EVALUATED.

6.2 Comparators and ablations

Dataset-specific frozen matrices give methods the same allowed target-normal support and descriptors. Learned methods share the encoded initialization; fixed templates use the frozen matched-normal control. All matrices include CADENCE plus linear, additive, and black-box learned methods with the same latent and hidden dimensions. No-effect, condition–time, and nearest-donor comparators are included where defined. Allen additionally includes a cell-functional ridge atlas and two CADENCE ablations. Teacher uses its separately listed native-decoder and adaptation/residual diagnostics. Frozen ICMS v1 emits no CADENCE ablations; a criterion requiring one would remain NOT EVALUATED. A learned pooled neural model without animal maps is unavailable because heterogeneous neuron counts do not define a common output dimension. Evaluation-only oracles are excluded from the baseline envelope. The reporting layer constructs a separate post-outcome envelope for each target and endpoint: the maximum observed score over *all* eligible non-oracle

comparators and ablations. This deliberately conservative reference is not a validation-selected single baseline and is not a deployable model-selection rule.

6.3 Uncertainty and animal-level inference

A positive claim requires predictive uncertainty that integrates donor-bootstrap ensembles, bootstraps of target normal support, donor intervention random effects, and process variability estimated without target interventions. It also requires split-conformal calibration using whole-animal cases disjoint from fitting, early stopping, and model selection (or an equivalent predeclared cross-fitted construction). Maximum standardized trajectory residuals form a simultaneous 90% band, with pointwise coverage, whole-trajectory coverage, and width retained. The 95% lower bound for whole-trajectory coverage is the exact one-sided binomial bound across independent animals, not a percentile bootstrap over binary indicators. In particular, 5/5 covered ICMS animals cannot certify a lower bound of 0.80. A runner that omits the required draws, disjoint calibration provenance, bands, or coverage fields cannot satisfy the corresponding gate; reporting fails closed to NOT EVALUATED.

The primary mean and gain intervals are 95% nonparametric bootstraps over animals (20,000 resamples). Paired animal-level sign-flip tests are exhaustive; sessions, trials, and neurons are never permuted as if independent animals. For the teacher benchmark, four targets are first averaged within world and the 20 worlds are bootstrapped or sign-flipped. These intervals are procedural only; the teacher conjunction remains NOT EVALUATED. Duplicating targets within a world cannot tighten inference. Incomplete biological headline cohorts cannot evaluate a gate.

6.4 Falsification and leakage audits

Gate 6 jointly requires four exact paired animal-level sign-flip tests (neural and behavior skill above zero, and neural and behavior gain above zero) plus three explicit randomization controls: target-label permutation, donor intervention-semantic shuffle, and animal-adapter shuffle. Gate 8 consumes pseudo-onsets on normal trials and a pre-onset effect check. Each randomization control retains its frozen procedure, test value, and artifact. Its canonical relative filename and observed digest must be bound into the runner’s authenticated completion chain; a bare pass flag or claimed digest is insufficient. Each negative control retains an equivalence margin and confidence interval whose bounds must lie inside that margin. Failure to reject a conventional null is not silently treated as evidence of equivalence: pre-onset and pseudo-onset controls require their explicit null criteria. A runner that omits a required diagnostic leaves the associated gate NOT EVALUATED. Onset shifts, initial-state-only decoding, and the teacher impossibility pair are additional diagnostics but are not silently counted as one of the declared gate inputs; frozen v1 does not emit all of them. Sentinel invariance is an executable leakage test rather than a biological null. A changed prediction hash, a target intervention row in a fitted transform, or a sealed sentinel reaching a trainable module invalidates the fold.

6.5 Conjunctive headline

For an eligible biological dataset, the overall status is PASS only when all eight rows pass. FAIL is retained even if another row is missing; otherwise any missing component yields NOT EVALUATED. Allen development mice never enter the headline. ICMS83 never enters its randomized causal summary. Teacher results test recoverability under known assumptions but cannot establish a biological shared operator. Because the teacher seeds are public, reporting fixes its headline conjunction to NOT EVALUATED regardless of numerical completeness. Frozen v1 is fail-closed: its

Table 2: **Pre-outcome conjunction ledger.** The statuses below are intentionally fixed to NOT EVALUATED until complete frozen cohorts are scored. A single failed gate defeats the corresponding dataset-level headline. The teacher benchmark is reported separately and cannot establish a biological claim; its public seeds make it ineligible for this conjunction.

Gate	Frozen criterion	Allen	ICMS
1	Randomized manipulation has nonzero neural and primary-behavior effects.	NOT EVALUATED	NOT EVALUATED
2	Equal-animal 95% CI for neural causal skill lies above zero.	NOT EVALUATED	NOT EVALUATED
3	Equal-animal 95% CI for primary-behavior causal skill lies above zero.	NOT EVALUATED	NOT EVALUATED
4	Mean gain is at least 0.10 over the per-target post-outcome envelope of all eligible non-oracle comparators and ablations for both endpoints, and both gain CIs lie above zero.	NOT EVALUATED	NOT EVALUATED
5	State-conditioned prediction improves proper score over the donor condition-time template.	NOT EVALUATED	NOT EVALUATED
6	Four exact paired animal-level sign-flip tests and all three declared animal/action randomization controls reject their nulls.	NOT EVALUATED	NOT EVALUATED
7	The 90% simultaneous bands have an exact one-sided 95% binomial coverage lower bound of at least 0.80 across animals.	NOT EVALUATED	NOT EVALUATED
8	Pre-onset and pseudo-onset controls satisfy their explicit null checks.	NOT EVALUATED	NOT EVALUATED

current biological runners do not emit everything needed to pass Gate 1 or Gates 5–8. Missing components remain NOT EVALUATED, while an authenticated evaluated subcomponent that contradicts a composite gate can still make that gate FAIL. This release can therefore return FAIL, but it cannot return a biological PASS. Missing evidence cannot be retrofitted after unsealing.

7 Pre-outcome development record

The development sandbox is reported only to disclose method selection and the known risk, not as confirmatory evidence. After removing validation-animal normal leakage, ten teacher smoke worlds gave proposed condition-averaged neural skill mean 0.00173, median 0.0131, range $[-0.1175, 0.0623]$, and 8/10 positive worlds. Proposed neural pathwise skill had mean 0.00878 and was positive in 7/10. Behavior skill was positive in 10/10 (mean 0.5049), whereas latent-effect skill was positive in 7/10 (mean 0.1270). The evaluation-only gauge-plus-true-observation diagnostic had mean neural skill 0.0278. On average, 9.79% of readout query coordinates fell outside the coordinate-wise range of target normal rollouts. This mixed, near-zero neural result does not establish reliable latent, operator, or observation-map transfer; it exposes estimation failure and a measurable off-range burden consistent with, but not diagnostic of, off-support extrapolation. Across the four allowed Allen development mice, proposed neural causal skill had mean -8.63×10^{-5} , range $[-1.85 \times 10^{-4}, 1.61 \times 10^{-7}]$, and 1/4 positive mice. Proposed running causal skill had mean -4.91×10^{-4} , range $[-1.69 \times 10^{-3}, 1.94 \times 10^{-4}]$, and 2/4 positive mice. Thus neither Allen endpoint established transfer.

No held-out ICMS stimulation outcome was used for model selection. A pre-freeze schema regression did load one ICMS83 example container to verify exceptional channel wiring, dimensions, catch count, and normal-ITI validity; it did not print, score, summarize, or tune against a response trajectory. ICMS83 is catch-free and excluded from the randomized causal cohort, and none of the five randomized-effect mice was opened. These observations and accesses were recorded before procedural evaluation outputs were run or randomized biological outcomes were unsealed. They cannot be overwritten by choosing a new architecture after seeing evaluation outcomes.

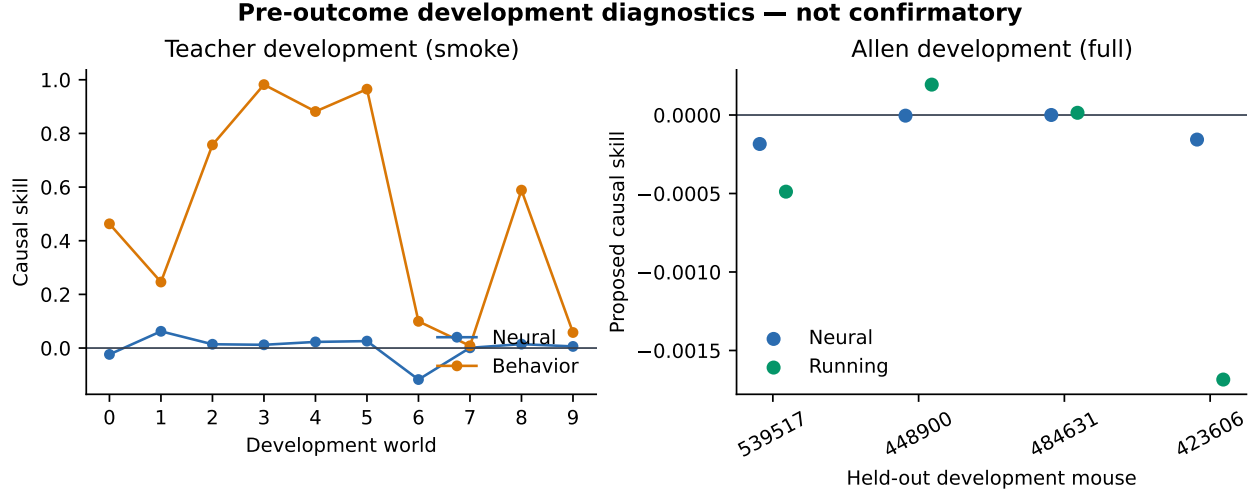


Figure 2: **Authenticated pre-outcome development diagnostics.** Left: proposed condition-averaged neural and behavior causal skill in the ten teacher smoke worlds. Right: proposed neural and running causal skill in the four permitted Allen full-development mice. These observations disclose method-selection risk and are not confirmatory evidence.

8 Prospective results

Prospective result slot — not populated. Teacher predeclared procedural evaluation (public seed; non-headline).

Insert all 80 target predictions, nested four per each of 20 independent deterministic post-freeze worlds; average targets within world before inference. Report observed-space condition-averaged neural and behavioral skills; realized-path diagnostics; latent/operator recovery only after a normal-fit gauge; any separately instantiated diagnostic stress slices; impossibility-pair behavior only if separately instantiated (otherwise NOT EVALUATED); all baselines; and a complete audit. Disclose the public seeds, state explicitly whether neural observation-map transfer succeeds, and retain the conjunction as NOT EVALUATED.

This box specifies planned reporting content. It contains no evaluation outcome and makes no performance claim. It may be replaced only from the immutable post-freeze aggregate and, for outcome-sequestered biological data, only after authorized unsealing, with failed and unevaluated gates retained verbatim.

Prospective result slot — not populated. Allen primary evaluation.

Insert one row per each of the 28 evaluation mice and the equal-animal aggregate. Report neural event-rate and running causal skill first, strongest baseline envelopes and uncertainty next, then pupil/lick secondary analyses, matching fallback distributions, proper scores, coverage, falsifications, and all eight gate statuses. Do not use secondary behavior to rescue running.

This box specifies planned reporting content. It contains no evaluation outcome and makes no performance claim. It may be replaced only from the immutable post-freeze aggregate and, for outcome-sequestered biological data, only after authorized unsealing, with failed and unevaluated gates retained verbatim.

Prospective result slot — not populated. ICMS exploratory direct intervention.

Insert all six absolute-trajectory results, but restrict the randomized causal aggregate to ICMS92, ICMS93, ICMS98, ICMS100, and ICMS101 and label $n = 5$. Report primary artifact-robust sorted-spike and wheel trajectories, physical current/site strata, uncertainty, and every unevaluated field. Keep longer/noncanonical train rows `OUT_OF_SCOPE_NOT_SCORED` and their analysis `NOT EVALUATED`; do not score them as a frozen-v1 sensitivity. Explicitly retain full-train spike and sparse-calcium analyses as `NOT EVALUATED` in frozen v1. ICMS83 must remain absolute-only.

This box specifies planned reporting content. It contains no evaluation outcome and makes no performance claim. It may be replaced only from the immutable post-freeze aggregate and, for outcome-sequestered biological data, only after authorized unsealing, with failed and unevaluated gates retained verbatim.

Prospective result slot — not populated. Conjunction and claim sentence.

Insert the machine-generated conjunction table without manual relabeling. Write a positive claim only for a dataset whose eight gates all pass. If a required gate fails, lead with the failure. If no gate fails but required evidence is missing, use “not evaluated,” not “passed” or “inconclusive success.”

This box specifies planned reporting content. It contains no evaluation outcome and makes no performance claim. It may be replaced only from the immutable post-freeze aggregate and, for outcome-sequestered biological data, only after authorized unsealing, with failed and unevaluated gates retained verbatim.

9 Interpretation

The frozen design supports three qualitatively different outcomes. First, positive neural and behavioral transfer with calibrated uncertainty would support a shared operator over the measured animal, state-support, and intervention family. It would not imply globally novel-action generalization or universal neural coordinates. Second, transferable behavior but failed cell-resolved neural prediction would be consistent with a shared coarse latent effect whose target observation map is non-identifiable off normal support. Third, failure already in latent or behavioral teacher endpoints would implicate dynamics/operator estimation rather than only the biological observation map.

Negative evidence is scientifically informative here. A model can reconstruct normal data, align animals, and even recover a donor perturbation field while failing the target response. Such a result limits what observational cross-animal alignment licenses us to infer. The protocol therefore commits to publishing failed and unevaluated gates rather than treating the locked cohort as a second development set.

10 Limitations

1. **Identifiability is assumption-dependent.** Low-rank structure, donor conservation, and normal-state coverage are inductive biases, not proofs that a target observation map extrapolates correctly.
2. **Interventions are not globally novel.** The target action is unseen in that animal but represented in donors. Extrapolation to a new action class remains untested.

3. **Allen scope is narrow.** The primary cohort is one cortical area, excitatory cell class, depth, task familiarity, and omission family. Matching estimates an eligible-risk-set effect and does not create paired counterfactuals.
4. **ICMS is small and exploratory.** Only five animals support the randomized causal contrast; ICMS83 is nonrandomized for effects. The associated bioRxiv v1 manuscript is accepted at *Science Advances*, but no version-of-record DOI was located as of 2026-07-25; stimulation artifact, longitudinal learning, and implant geometry limit transport.
5. **Modalities differ in support.** Sorted spikes are primary for ICMS. Volumetric calcium is sparsely observed by z-plane and cannot be interpreted as a simultaneous 30-Hz population.
6. **Open-loop error compounds.** Long-horizon failures can combine initialization, normal-flow, intervention-field, and observation-map error. Oracle diagnostics localize failure but are not eligible evidence.
7. **Uncertainty can remain underpowered.** Animal-level inference correctly exposes rather than hides the small biological sample, particularly for ICMS.

11 Reproducibility, data governance, and ethics

Raw public datasets are excluded from version control; committed manifests record immutable versions, asset identifiers, byte counts, and content hashes. Locked profiles require a clean checkout at the exact annotated `pre-outcome-v1.0.0` tag. Predictions and outcome bundles are physically separate, and every prediction archive receives a SHA-256 digest before scoring. Complete frozen aggregation uses the predeclared 20,000-resample seed and runs only from the clean tagged reporter; locked input paths and SHA-256 digests are retained. The final machine-readable summary, long-form CSV, compact L^AT_EX table, and conjunction ledger are published atomically into a new append-only directory with artifact sidecars and a completion manifest. Missing uncertainty, coverage, or falsification fields remain NOT EVALUATED.

This project performs secondary analysis of public animal datasets and a procedural simulation; it introduces no new animal experiment. Ethical and regulatory approvals for data acquisition are documented by the source studies. Reuse is subject to the source licenses and requires citation of the Allen release and DANDI:001868.

12 Conclusion

CADENCE is a prospective biological falsification test, not a guarantee that causal dynamics transfer across animals. Its scientific value rests on a strict information boundary, observed-space endpoints, animal-level replication, and a decision rule that is allowed to fail. The latent gauge is harmless for observed prediction; extrapolation of a target neural map beyond normal support is not. The frozen biological evaluations and public-seed procedural teacher audit will determine whether the imposed structure is sufficient in their declared scopes. Until then, the primary biological claim remains prospective.

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